

Prem Khanvilkar

Biomedical Engineering Student he/him

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TECHNICAL SKILLS

Programming

- Python
- MATLAB
- R
- C & C#

Laboratory & Hardware

- Circuit design/Analysis
- Oscilloscope
- Breadboarding
- Microcontrollers (Arduino)

Engineering Software

- SolidWorks
- Pymol
- SwissDock
- BLAST

EDUCATION

University of British Columbia

Bachelor of Applied Science - Biomedical Engineering

Expected Graduation: May 2029

WORK EXPERIENCE

Biochemical Innovation Technology, Vancouver, Canada

October 2025 - Present

Student Researcher

- Shadowed islet picking and β -cell extraction from primary human tissues and prepared experimental media using serial dilutions of polyphenols
- Quantify insulin secretion using ELISA assays and analyse results, generate and interpret experimental data using GraphPad Prism and Excel
- Conduct literature review and contribute to scientific report writing, and present findings during weekly meetings
- Present research at conferences such as Multidisciplinary Undergraduate Research Conference (MURC)

The Fourth Perspective, YouTube Channel, Remote

March 2025 - Present

Co-founder

- Scaled a YouTube Channel nearing 9000 subscribers in under 4 weeks and accumulated over 200,000 views. Produced content about Engineering Lifestyle and guided students to apply to international universities

Bifrost EdTech, Delhi, India

March 2023 - November 2023

Chief Human Resource Officer

- Led a professional HR Team, overseeing over 1000 students in company across different departments, with no reported grievances
- Drafted legal documents such as Memorandums of Understanding, Codes of Conduct and Offer Letters. Responsible for distributing certificates and recommendation letters for over 250 students
- Administered outreach and hiring an efficient workforce for every department, conducting over 50 interviews for board and general members

RESEARCH & TECHNICAL PROJECTS

Biomechanical EMG Sensor Analysis of Bicep Curls, UBC

November 2025

- Configured surface electromyography (EMG) electrodes and a BioRadio system to measure activation signals of bicep across varying arm and shoulder orientation during bicep curls
- Filtered raw EMG signals to extract linear envelope, processing data to calculate root mean square (RMS), peak, and mean activation values

Orthopaedic Implant Mechanical Analysis, UBC

October 2025

- Engineered fracture fixation plate models and conducted static simulation using SolidWorks to evaluate structural integrity and load-bearing capacities under peak physical stresses



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- Analysed and compared mechanical performance of Titanium-6Al-4V against Alloy Steel and Aluminium, and concluded minimising screw hole diameter directly reduces stress concentrations and increases overall hardware resilience

Molecular Docking & Biochemical Analysis, UBC

January 2026 - April 2026

- Evaluated protein models (AlphaFold) against experimental X-ray crystallography structures, and selected most reliable models based on sequence alignment
- Processed complex 3D receptor coordinates in PyMOL by isolating target protein chains and preserving essential cofactors to ensure accurate geometry
- Executed molecular docking simulations using SwissDock to evaluate thermodynamic data of various compounds. Converted thermodynamic energy outputs into dissociation constants to correlate docking scores directly with experimental lab data

Grip Strength & EMG Force Correlation Analysis, UBC

November 2025

- Recorded surface EMG and hand dynamometer force data simultaneously using a BioRadio system at 2000 Hz during grip strength
- Processed EMG signals through a filter pipeline (bandpass, notch, rectification, envelope) and applied threshold-based onset detection to quantify the delay between muscle activation and force output
- Correlated EMG amplitude with grip force using linear regression to evaluate the relationship between muscle activation and generated force

Kinematic Motion Capture Analysis, UBC

November 2025

- Captured elbow and wrist kinematics during water-drinking motion using marker-based and marker-less tracking systems
- Applied FFT analysis to determine an appropriate cutoff frequency and designed a 4th-order zero-phase Butterworth low-pass filter to smooth kinematic data
- Calculated angular velocity and tangential velocity using a rigid-body forearm model, and compared tracking accuracy and left-right symmetry between methods

VOLUNTEER EXPERIENCE

The Helping Paw & Careiyat NGO, Mumbai, India

June 2023 - November 2023

Co-Founder, Volunteer

- Co-founded a local animal care initiative within India's largest student-led NGO, The Helping Hand
- Focused on raising awareness about animal welfare and addressing issues of stray animal sanitation
- Contributed over 100 hours under Careiyat NGO, conducting food and sanitary drives

AWARDS & CERTIFICATIONS

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| • Dean's List Fall 2025 & Winter 2026 | April 2026 |
| • Certified in Chemical Safety, Biosafety & Design Team Safety | October 2025 |
| • Gold Medalist, WAKO International Kickboxing | November 2022 |
| • Workplace Hazardous Material Information System Certification | October 2025 |



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